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A Component-Based Approach for Developing Actions in Non-Humanoid Elderly Care Robots

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Developing expressive motion for non-humanoid companion robots is a growing area of research in human–robot interaction. This study presents a hybrid design approach that integrates designer-led ideation with user-based empirical validation to develop physical actions for Lemmy, a two-wheeled companion robot for elderly users. The research followed a three-stage process: designer workshops produced an initial action catalogue; pairwise comparison testing with older adults identified preferred variants of four key motion components, speed, posture, path, and turn type, and validation studies compared complete actions composed from preferred, non-preferred, and random combinations. High-fidelity 3D animation was employed to enable rapid iteration, experimental control, and parallel progress while the physical prototype remained under development. Results from 25 elderly participants showed consistent preferences for dynamic posture, smooth turning, and moderate speed, confirming that component-level optimization enhances perceived comfort and naturalness. Crucially, the findings demonstrate that once user-preferred motion components are established, they can be recombined to generate new, user-aligned actions without exhaustive re-testing. The proposed approach thus offers a scalable, designer-accessible method for creating expressive motion in non-humanoid companion robots for eldercare contexts.

Keywords: *companion care robots; action design; non-humanoid robots; Human-Robot Interaction (HRI)*

1 Introduction

The rapid growth of aging populations worldwide has created unprecedented demand for assistive technologies that enable older adults to maintain independence and quality of life (Sawik et al., 2023). Companion robots represent a promising intervention, capable of providing both functional assistance and social-emotional support to elderly users living at home (Macis et al., 2022). However, current companion robot research predominantly focuses on humanoid or zoomorphic designs, leaving non-humanoid platforms underexplored despite their potential advantages in cost, reliability, and functional flexibility (Hoggenmueller et al., 2020).



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Non-humanoid companion robots face a fundamental communication challenge: without anthropomorphic features, these platforms must convey intent, emotion, and social presence through movement alone (Xie & Luh, 2023). Research has established that motion characteristics influence user perception of robot affective states and intentions. Yet movement expressivity in non-humanoid forms remains inadequately understood, particularly within eldercare contexts where user acceptance hinges on perceived safety and predictability (Scales et al., 2024).

Existing approaches to robot motion design reveal three critical gaps. First, theoretical frameworks describing expressive motion rarely translate into systematic, practitioner-accessible design methodologies (Hoffman & Ju, 2014). Second, participatory design methods involving elderly end-users encounter practical barriers: cognitive load, fatigue, and limited technical fluency make intensive co-creation sessions challenging (Sinner & Lim, 2021). Third, no validated methodology exists for decomposing robot actions into designable components that can be systematically optimized and recombined (Hoggenmueller et al., 2020).

These gaps create a methodological impasse: designers lack structured processes to develop expressive non-humanoid motion, while direct participatory approaches with elderly users prove impractical for iterative exploration (Sinner & Lim, 2021). Consequently, non-humanoid companion robot motion often emerges from ad hoc intuition rather than empirical validation, risking misalignment with older users' comfort thresholds and social expectations.

Lemmy, the subject of this study, is a two-wheeled, non-humanoid companion robot designed to support older adults in their own homes (see Figure 1). Instead of a humanlike form, Lemmy conveys emotion and intent through gentle movements, on-screen animations, and soft sounds. It offers reminders, emotional reassurance, and a comforting presence, all crafted to promote well-being without overtly anthropomorphic features. Lemmy self-balances on the two wheels and can lean forward and backward by 20° . Lemmy has differential drive which means Lemmy is a non-holonomic system and supports neutral steering (pivot turning). This research fills a gap by exploring how Lemmy's actions can be designed and evaluated for optimal eldercare support.

This study introduces a component-based hybrid methodology that systematically integrates designer expertise with user-driven validation. The approach decomposes robot actions into four interpretable motion components each independently optimized through low-cognitive-load pairwise comparison testing with elderly participants. Complete actions are then synthesized from empirically validated components, balancing creative design exploration with rigorous user validation while minimizing participant burden.

The research examines whether component-level optimization yields actions perceived as more natural and comfortable than non-optimized alternatives and tests the hypothesis that validated components can be recombined to generate novel actions without exhaustive re-validation. By establishing this component-to-action pipeline, the methodology offers a scalable, reproducible framework for developing expressive motion in non-humanoid eldercare robots, transforming action design into an evidence-based practice accessible to designers without specialized robotics expertise.



Figure 1. Lemmy - Companion Care Robot

2 Related work

2.1 Companion care robot design for elderly users

The landscape of companion robot research has evolved significantly over the past decade, with increasing attention to the specific needs and preferences of elderly users. Early studies demonstrated that companion robots such as Paro could reduce agitation, depression, and loneliness among older adults while decreasing caregiver burden (Shibata & Wada, 2011). However, these initial investigations primarily focused on predetermined robot designs without extensive user input during the development process, leading to mixed results and limited real-world adoption (Hung et al., 2019).

Recent research has emphasized the critical importance of user-centered design in developing companion robots for elderly populations. Bradwell's comprehensive investigation revealed significant differences between end-users and developers in perceptions of suitable robot design, with older adults preferring soft, furry, familiar, and realistic embodiments over the abstract designs favored by roboticists (Bradwell et al., 2019). This finding underscores the necessity of involving elderly users throughout the design process rather than relying solely on expert assumptions about user preferences. Older adults also demonstrate stronger preferences for functional, service-oriented robots that provide tangible assistance rather than purely social interaction. The appearance of a humanoid robot is often described as favorable, but the definition of 'humanoid' varies widely within all analyzed studies, and an explicit allocation of features is not always defined (Chiu et al., 2021). Preferences are not uniform; some older adults value less interactive, less intrusive robots, while others appreciate more expressive, social designs (Coghlan et al., 2021). Cross-cultural studies have further revealed that design preferences may vary across different populations, though some fundamental principles appear to transcend cultural boundaries (Avijeet et al., 2024; Zhang et al., 2021). These findings highlight the need for participatory methods in designing companion robots for the elderly.

While these studies highlight the necessity of user-centered design, they also expose practical constraints in involving elderly users throughout early-stage development. These limitations motivate a structured, designer-led approach that retains user input without overburdening participants.

2.2 Action design and movement characteristics in non-humanoid robots

The design of robot actions and movements is a widely studied aspect of companion robot development. Traditional approaches to robot behavior design have often focused on mimicking human or animal movements, but recent research suggests that non-humanoid robots may require fundamentally different approaches to action design (Cass et al., 2018). The challenge lies in creating expressive and meaningful movements within the constraints of non-anthropomorphic forms while maintaining user engagement and understanding.

Movement characteristics such as speed, path, and posture have been identified as key parameters that influence user perception and acceptance of robotic systems. Users interpret movement qualities as indicators of intent, mood, and internal state, even in robots without human-like forms. The expression patterns of emotional arousal through motion amplitude are roughly the same across different robots: the degree of motion amplitude corresponds closely to the degree of arousal (Xie et al., 2025). Research on robot movement speed has demonstrated that optimal speeds vary depending on context, user demographics, and intended functionality. Moderate speeds are generally preferred for efficiency and engagement, but slower, more predictable movements are valued for safety - especially when users are unfamiliar with the robot or in sensitive contexts (Gervasi et al., 2022). Studies with elderly users specifically have shown preferences for slower, more deliberate movements that convey safety and predictability while avoiding the perception of weakness or incapacity. Users often anthropomorphize robots based on movement, attributing emotions and agency even to non-humanoid forms (Law et al., 2021).

Path planning and navigation strategies also significantly impact user comfort and acceptance. Research has indicated that curved, pet-like movement paths may be perceived more favorably than rigid, mechanical trajectories, particularly in domestic environments. However, the right methodologies to find the optimal balance between natural movement and functional efficiency remain an important concern for non-humanoid companion robots that lack clear biological analogs for movement behavior. These insights highlight the importance of decomposing robot motion into distinct, design-controllable parameters that can be systematically optimized through user testing.

2.3 Identified Gaps in Designing and Evaluating Actions for Non-Humanoid Robots

Although HRI has matured in methods for co-design and evaluation, there remains a methodological gap in the granularity of individual robot actions for non-humanoid platforms in eldercare. Engagement with older adults is essential yet difficult to sustain at the cadence required for fine-grained motion tuning; empirical reviews of participatory HRI with older adults document fatigue, session burden, and constraints that undermine repeated exposure to micro-variants of movement, even when studies are well designed (Rogers et al., 2022). Recent work in aging-focused co-design similarly notes the practical limits of sustained participation, calling for approaches that reduce cognitive and physical load without sacrificing user voice (Olatunji et al., 2024). In practice, these constraints encourage teams to defer motion specification until late stages when a physical prototype is available, coupling expressive decisions to hardware readiness and conflating user preference with what the robot currently affords.

A parallel stream attempts to circumvent hardware bottlenecks via video or simulation, but the field still lacks a validated pipeline that connects low-burden stimuli to action-level decisions. Emerging evidence shows that online or video-based HRI yields results broadly comparable to lab settings for

several perceptual constructs, while also identifying dimensions where embodiment matters, thus motivating careful stimulus design rather than abandonment of pre-embodiment testing. More recently, controlled comparisons report systematic differences between interactive simulation and video for navigation competence judgments, underscoring the need to specify when and how screen-based methods are appropriate and how they should inform downstream embodied behavior (Tsoi et al., 2024). Meta-analytic work further clarifies where “not-so-real” robots approximate “real” ones and where they diverge, strengthening the case for principled pre-embodiment evaluation that feeds into embodied validation instead of replacing it (Esterwood et al., 2025). Yet these contributions stop short of prescribing how to decompose expressivity into reusable motion primitives that can be mixed, recombined, and selectively validated with older adults.

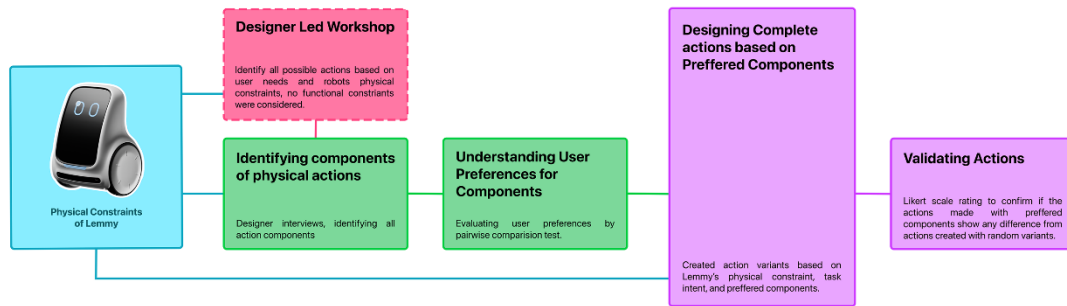
For non-humanoid robots specifically, perception studies show that users infer affect and intent from kinematic features rather than morphology, indicating that components of motion are the natural design levers for action synthesis (Law et al., 2021; Scales et al., 2024; Xie & Luh, 2023). Likewise, trajectory-level modeling now links specific velocity features to social attributions and offers optimization handles for shaping perception, but this work remains largely engineer-facing and does not solve the designer’s need to elicit and generalize user preferences at the component level before embodiment (Scales et al., 2024). Finally, interview studies with robot creators repeatedly surface the translation problem: qualitative design intentions such as “approachable” or “attentive” are difficult to render as parameterized, physically consistent trajectories under safety and platform constraints, creating frictions at the handoff from design to engineering, frictions that are amplified in eldercare contexts where predictability and legibility are non-negotiable (Alves-Oliveira et al., 2022).

Taken together, the field lacks a designer-accessible pipeline that (i) minimizes participant burden for older adults, (ii) decouples action refinement from prototype availability while respecting known limits of screen-based studies, (iii) operationalizes non-humanoid expressivity as combinable motion components, and (iv) supports faithful translation from expressive intent to parameterized, constraint-respecting motion. The present work addresses this gap by structuring action design around validated components and by using high-fidelity animation to elicit preferences that can be recombined into new actions prior to embodied testing.

3 Methodology

We began from Lemmy’s finalized industrial design and explicitly bounded all actions by the platform’s physical and kinematic constraints (e.g., lean limits, non-holonomic drive). To iterate before hardware readiness, we built a high-fidelity digital twin: the parametric CAD model was exported to STL and imported into Blender, where a polygonal pipeline enabled precise, constraint-respecting animation and photorealistic rendering. Because each motion was authored within these feasibility limits, results from the 3D environment transfer directly to the functional prototype. On this foundation, we pursued a three-stage process, designer-led action generation, component-level user preference testing, and action-level validation, to turn early concepts into empirically preferred behaviours for a non-humanoid companion robot (Figure 2).

Figure 2. Overview of overall methodology



3.1 Action Catalogue

Following preparations we ran a designer-led workshop with six designers (three teams of two) to enumerate candidate robot actions, cluster them, and assign an initial hierarchy based on designer judgment. To keep ideation anchored in realistic use, teams worked from three detailed user stories reflecting typical routines of older adults. Designers were asked to record inferred robot actions without enforcing functional constraints; nevertheless, awareness of platform physical limitations informed their suggested actions.

This workshop produced three outputs: (1) a consolidated list of inferred actions, (2) their frequency of mention across teams, and (3) candidate clustering schemes. These outputs informed the construction of an action catalogue that is contextually grounded, physically feasible, and aligned with product expectations. The catalogue now serves as a working reference for scenario illustration, user testing, and systematic action development. To prioritize within clusters, we recorded whether an action was selected by all three teams or by a subset, using occurrence as a simple within-cluster salience signal. The workshop outputs (Figure 3) have been reported previously (Sarfraz et al., 2025).

##	Category	Sub-category	Actions	Occurance
1a	Locomotion and Navigation	Approaching & Following	Approach user	3
1b			Follow mode	3
1c			Return to dock/base	3
1d		Positional Adjustments	Repositioning / user tracking	3
1e			Obstacle avoidance	1
1f		Emergency Maneuvers	Retreat/reverse	1
1g			Navigate to help	1
2a	Expressive movements	Head & Upper Body	Nodding	3
2b			Head shaking	2
2c			Tilting/leaning	1
2d		Emotional Expressions	"Happy" wiggle	1
2e			"Sad" slump	2
2f			"Surprised" jolt	2
2g		Rhythmic/Dance	Dancing/swaying	1
2h			Task Completion pose	1
3a	Interaction and assistance	Object interaction	Dragging / pushing objects	2
3b			Object tracking	2
3c			Pointing / Indicating	1
3d		Guidance	Leading the way	1
3e			Signaling Path	1
4a	Extras	Self-Maintenance		
4b			Room scanning / Finding User	1
4c			Roam around house	1

Figure 3. Developed action catalogue for "Lemmy" companion care robot through designer-led workshop

3.2 Identifying Action Components

To enable structured and interpretable control over the robot's motion, we define action components as high-level, designer-specified parameters that capture salient variations in movement and can be systematically composed to form complete actions. This component-based representation mitigates combinatorial complexity while providing designers with meaningful, human-interpretable control dimensions that older adults can readily perceive and assess. By abstracting away from low-level control variables (e.g., yaw angle, wheel RPM) that offer limited perceptual or experiential value, this approach prioritizes motion features with direct relevance to human judgment and interaction quality. Evidence in HRI shows that people reliably read kinematics (acceleration, curvature, and path shape) as cues to affect, intention, and social stance even for non-humanoids; navigation surveys likewise tie speed, clearance, and trajectory selection to comfort and legibility in shared spaces (Kruse et al., 2013; Saerbeck & Bartneck, 2010). Laban-inspired feature sets for mobile robots offer a practical vocabulary (Space/Time/Weight/Flow) for authoring expressive motion on low-DOF platforms, and anticipatory motion increases observers' ability to infer intent earlier (Knight & Simmons, 2014). Form-and-motion studies further support prioritizing kinematic cues even when morphology is simple (Castro-González et al., 2016).

To enumerate components relevant to our platform, we ran semi-structured interviews with five external professional designers (five with >3 years' industry experience). Sessions (30–60 minutes) were conducted via Google Meet and introduced the robot and elder-care context before probing non-humanoid motion possibilities, user considerations, and the specific motion variables designers manipulate. This method surfaces tacit schemas that help translate qualitative intentions into parameterized, physically consistent motion and yields a practicable, designer-facing component vocabulary. Examples emerging as both feasible and interpretable include speed, posture/orientation, path shape, and turn type.

3.3 Evaluating user preference for action components

For Lemmy, action was decomposed into four components, speed, path, posture, and turn type, each implemented in two variants to allow clear, binary preference testing without overloading participants or introducing confounds. This simplified design enabled unambiguous pairwise comparisons while capturing salient behavioural contrasts, with finer gradations (e.g., subtle speed or curvature variations) intentionally omitted to preserve clarity and reduce fatigue in the elderly sample. Among these, posture was the most distinctive, as Lemmy could either maintain a static upright stance or adopt a dynamic action specific lean. Speed was tested at a normal level ($\approx 0.5\text{--}0.7$ m/s), reflecting the casual walking pace of a similarly scaled quadruped pet, and a faster condition (>0.7 m/s). The path dimension contrasted straight trajectories with sinusoidal, organically curved motions, while turn type was varied between sharp pivot turns and smooth, arc-like rotations around a corner.

Preferences were measured using a within-subjects paired-comparison experiment with twenty-five participants aged over sixty-five, all based in Pakistan. Participants viewed 3D animations of Lemmy performing paired variants of each component and indicated their preference on each trial. The within-subjects design reduced between-person variability, randomization controlled for sequence effects, and the resulting data identified which traits elderly users valued most.

All materials were presented as high-fidelity 3D animations of the robot's digital twin with controlled movement parameters and a uniform viewing procedure. This ensured that judgments reflected

differences between components rather than incidental presentation artifacts. The choice of pre-rendered animation was methodological, not merely convenient: it enabled rapid iteration of components, removed hardware confounds, and let us run user studies while the platform was under active development. This aligns with prior evidence that video/simulation prototypes are faster and cheaper for early evaluation and with HRI norms that validate pre-embodiment user testing.

3.4 Designing and validating individual actions

This phase synthesized the component-level findings into complete robot actions and tested whether those preferences scale. For each action, four variants were constructed: one assembled from preferred component, one from non-preferred, and two designed by random combinations. The study asked a single question: do actions composed of individually preferred components receive higher judgments than non-preferred or random assemblies? A new cohort of fifteen participants viewed twelve videos in total (three actions $\times$ four variants) under the same stimulus presentation setup as earlier.

Three catalogued actions were selected to validate the scalability: “approach user,” “returning to dock,” and a “dancing action.” The first two were prioritized for detailed analysis, as they are fundamental to companion-robot operation and exhibit variation across multiple movement dimensions. The dancing action, in contrast, was included to emphasize expressive qualities rather than task-oriented performance. During the creation of action variants, turn type, although initially assessed for user preference, was deliberately excluded, as it functions as a general aspect of translational motion rather than a feature specific to any single action.

Participants provided two judgments for every video: a 10-point Likert rating and brief open-ended feedback. The Likert scale captured graded evaluations suitable for quantitative analysis and comparison across variants. The short comments recorded participants’ stated reasons for those ratings, supporting interpretation of quantitative patterns and revealing cues not fully conveyed by component labels. Pairing a bounded numeric scale with concise free-text responses balances simplicity for an elderly cohort with sufficient diagnostic depth for design decisions.

The resulting ratings and comments provided an action-level validation of the component approach and supplied concrete guidance for constructing preferred action sequences in subsequent development.

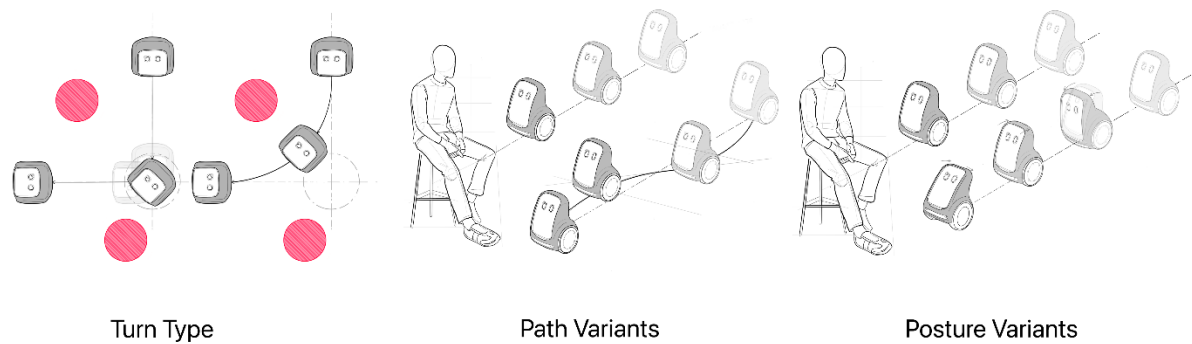
4 Results

4.1 Interview Results

Thematic analysis of the interview data revealed three overarching themes that describe participants’ perceptions of the robot’s movement qualities. Most participants began by identifying all the moving parts and considering the larger unique components that could be derived from them. For instance, in relation to translation, participants most mentioned speed and path. Upon further questioning, the concept of path was divided into two aspects: the path followed while traveling, referred to in this study as “Path,” and the movement involved in turning around a corner (Turn Type). In addition, due to Lemmy’s unique self-balancing feature, participants identified leaning as an additional variable. Other components that emerged from the discussions included rhythm or repetition, as well as acceleration. In the end only four: speed, path, posture, and turn type were selected for component

preferences (Figure 4). These were selected for systematic evaluation because they appeared frequently in the interviews and were interpretable by designers and end users, ensuring that subsequent testing addressed dimensions that were both actionable in design and perceivable in practice.

Figure 4. Three of the four repeated action variants



4.2 Component identification and user preferences

Across 25 observations the pattern is apparent (Table 1). Dynamic posture was chosen in 16/25 cases (64%), straight paths in 17/25 (68%), arc/smooth turns in 16/25 (64%), and normal speed in 20/25 (80%). These consistent majorities define a clear default configuration: dynamic posture, straight path, arc/smooth turning, and normal speed.

This configuration operationalizes the method's baseline. It maximizes perceived legibility and comfort while preserving expressive headroom. We adopt these proportions as priors for action synthesis and as add-on defaults; departures from this set are intentional design moves to encode specific affect or emphasis.

Table 1. Participant preferences by component ($n = 25$).

Traits	Variant 1	Variant 2
Speed	Fast (20%)	Normal (80%)
Posture	Static (36%)	Dynamic (64%)
Path	Straight (68%)	Curved (32%)
Turn Type	Sharp (36%)	Arc (64%)

4.3 Validating individual actions

Participants ($n = 15$) rated four variants of each action (V01 = assembly of preferred components; V02 = assembly of non-preferred components; V03–V04 = random recombination). We report mean (M), standard deviation (SD), two-sided 95% confidence intervals for the mean (CI; Student's t , $df = 14$), and median (Md). Table 2 provides both, average per action as well as combined average of each variant.

Table 2. Within-subject ratings by action and variant (n=15).

Action Variants	Components	Approach User	Return to dock	Dancing	Combined Average
Variant 01 (V01 -Preferred)	Dynamic Posture, Straight Path, Normal Speed	7.5	8.5	7	7.7
Variant 02 (V02 - Non-preferred)	Static Posture, Curve Path, Fast Speed	5.7	5.4	4.9	5.3
Variant 03 (V03 - Random)	Dynamic Posture, Curve Path, Fast Speed	6.6	5.7	6.7	6.3
Variant 04 (V04 - Random)	Static Posture, Straight Path, Normal Speed	5.7	6.7	6.3	6.2

For Action 01 (Approach User), V01 received the highest evaluations ($M = 7.53$, $SD = 1.64$, 95% CI [6.62, 8.44], $Md = 8$). V03 followed ($M = 6.60$, $SD = 2.29$, 95% CI [5.33, 7.87], $Md = 7$), whereas V02 and V04 were lower and comparable (V02: $M = 5.67$, $SD = 1.29$, 95% CI [4.95, 6.38], $Md = 6$; V04: $M = 5.67$, $SD = 2.29$, 95% CI [4.40, 6.93], $Md = 5$). Dispersion was notably larger for V03 and V04 ($SD \approx 2.29$), indicating greater heterogeneity in responses, while V02 showed the tightest agreement ($SD = 1.29$).

For Action 02 (Return to Dock), V01 was clearly preferred, combining the highest central tendency with the smallest spread ($M = 8.47$, $SD = 0.88$, 95% CI [7.98, 8.95], $Md = 9$). V04 formed a middle tier ($M = 6.67$, $SD = 1.64$, 95% CI [5.76, 7.57], $Md = 8$). V02 and V03 were lower and more variable (V02: $M = 5.40$, $SD = 1.99$, 95% CI [4.30, 6.50], $Md = 6$; V03: $M = 5.67$, $SD = 2.56$, 95% CI [4.25, 7.08], $Md = 6$), with V03 showing the greatest polarization ($SD = 2.56$). The CI for V01 does not overlap with any other variant in this action, underscoring its descriptive advantage.

For Action 03 (Dancing), V01 again led ($M = 7.00$, $SD = 1.34$, 95% CI [6.26, 7.74], $Md = 7$), closely followed by V03 ($M = 6.73$, $SD = 1.65$, 95% CI [5.82, 7.65], $Md = 7$). V04 occupied a moderate position ($M = 6.27$, $SD = 1.69$, 95% CI [5.33, 7.20], $Md = 6$), and V02 was the weakest ($M = 4.93$, $SD = 1.94$, 95% CI [3.86, 6.01], $Md = 5$).

Overall (collapsed across actions; 45 ratings per variant), the ranking was unambiguous: V01 was highest ($M = 7.67$, $SD = 1.43$, 95% CI [7.24, 8.10]), followed by V03 ($M = 6.33$, $SD = 2.20$, 95% CI [5.67, 6.99]), V04 ($M = 6.20$, $SD = 1.90$, 95% CI [5.63, 6.77]), and V02 ($M = 5.33$, $SD = 1.76$, 95% CI [4.81, 5.86]). Thus, assembling actions from preferred components (V01) yielded the most favourable and most consensual ratings across all actions; the non-preferred assembly (V02) was consistently least favoured; and the random recombination (V03–V04) occupied the middle, sometimes approaching V01 in mean ratings but with looser agreement.

5 Discussions

Taken together, these studies provide preliminary evidence that component-level preferences can inform whole-action design for a non-humanoid eldercare robot. In a small within-subjects' validation, actions assembled from component settings aligned with these preferences received higher descriptive ratings than non-preferred assemblies across three tasks. Because preferences for posture, path, and turn type did not reach significance in our initial sample, and because action-level inference

requires a repeated-measures model, we present these findings as directional and to be confirmed in larger, embodied studies.

Among the three actions, the “Return to dock” variant built from preferred components is the best overall. It not only achieves the highest evaluations but also attracts the tightest agreement, an indication that component-tuned legibility matters most where the goal state is obvious and safety-relevant. “Approach user” also benefits from the same component recipe but shows slightly more dispersion, consistent with the fact that approach behaviors carry social meaning and invite individual differences in comfort zones. “Dancing” elicits the widest spread of opinions, suggesting that when an action is explicitly expressive rather than task-oriented, users tolerate (and some prefer) departures from the restrained baseline. These contrasts reinforce a useful distinction: components like straight paths and arc-turns are especially valuable for task actions where predictability and clearance matter, while expressive actions can selectively relax those constraints without universally harming acceptance.

The main hypothesis that actions composed of individually preferred components would outperform non-preferred and ad-hoc combinations is validated. Differences are not only directional but large in practical terms: across actions, the preferred-component assemblies separate clearly from the non-preferred versions, and the two “random” recombination sit in the middle with higher variance. In other words, the component priorities learned from quick, low-burden comparisons are strong enough to be indicative which whole actions will be judged better by new participants who have never seen the component tests. This is the methodological contribution: preference measurement at the component level is not merely descriptive; it is indicative at the action level.

These findings reframe how expressive movement for non-humanoid companion robots can be designed. Rather than iterating on entire actions until something “feels right,” teams can first establish a small library of user-preferred component values for a given platform and context, treat those as design priors, and then compose new actions by combining them. The effect is a step-change in efficiency and reproducibility: (i) it reduces participant burden (critical in eldercare) because we avoid re-testing near-duplicates of the same kinematic ideas; (ii) it decouples early motion design from hardware availability by using high-fidelity animation as an experimental testbed; and (iii) it yields actions that are both legible and comfortable by construction. The practical upshot is that a modest amount of careful preference elicitation buys a great deal of downstream generalization.

We therefore suggest a component-first pipeline for non-humanoid action design. Begin by fixing the scope and physical constraints of the target robot and domain so that the component vocabulary is grounded in what the platform can do. Use designer-led workshops to enumerate and cluster candidate actions without prematurely over-fitting to any single use case. Derive a small set of interpretable movement components (Lemmy’s case: speed, posture, path, and turn type) through expert interviews, then elicit preferences for each component via low-cognitive-load paired comparisons with representative older adults. Consolidate the resulting “component priors” into a living library and use them to synthesize complete actions for core tasks; only when an action intentionally departs from the priors (e.g., for expressive purposes) is an additional check warranted. Finally, translate these specifications to the physical robot, validate a minimal set of action exemplars in-situ, and push any systematic discrepancies back into the component library so that the priors evolve with context and embodiment.

The implications for the field are two-fold. Substantively, the work shows that natural, comfortable behavior in non-humanoid eldercare robots does not require anthropomorphic mimicry; it emerges from calibrated kinematics that emphasize legibility. Methodologically, it demonstrates that designer-accessible, component-level testing can stand in for heavy participatory co-creation while still keeping older adults' preferences at the center. The impact is a replicable, scalable path from design intent to parameterized motion: a pipeline that other teams can adopt to expand action catalogues quickly, justify choices empirically, and reserve scarce user time for validating genuinely novel or boundary-pushing behaviors.

6 Limitations and Future Directions

Although the component-first approach demonstrated here offers a pragmatic route to user preferred motion design, its evidential scope is constrained by the morphology of the target platform. Lemmy lacks complex, articulated moving parts; the few dynamic elements available (most notably "posture") are limited in amplitude and resolution, which narrows the repertoire of legible cues and may attenuate effect sizes when translated from animation to hardware. In addition, we operationalized each component as a binary contrast to minimize cognitive load for older participants. While defensible methodologically, dichotomization can compress multiple latent constructs into a single label and leave residual ambiguity about what, precisely, respondents preferred (e.g., a "dynamic posture" may be read as attentional orientation, approach intent, or mere motion). Designers adopting these results should therefore treat the component priors as directional rather than prescriptive and, where design decisions hinge on fine distinctions, employ graded manipulations with manipulation checks to disambiguate user interpretations before locking specifications into the action library.

Our use of high-fidelity 3D animation delivered clear benefits, rapid iteration, fine experimental control, and decoupling from prototype readiness, but necessarily omitted interactional contingencies that shape perception in situ: mutual timing, proxemic negotiation, ephemeral collision-avoidance micro-adjustments, and reciprocal adaptation to human movement. As such, the present findings should be complemented with user-in-the-loop evaluations on augmented-reality (AR) or mixed-reality testbeds that render life-scale, physics-aware robot avatars in real environments. AR can probe comfort margins during approach, behavior near furniture, and occlusion-rich contexts that static video cannot capture, while preserving the speed and parallelism that make animation attractive in early phases.

A related methodological caveat concerns physical realizability. Animation-based pipelines implicitly assume that depicted motions are implementable on a differentially driven, nonholonomic base. If trajectories require lateral slip, violate curvature constraints, or exceed actuation envelopes for velocity, acceleration, or jerk, preference data become, at best, weakly actionable and, at worst, misleading. Future iterations should embed an engineering guardrail within the authoring workflow: a validated dynamics model of Lemmy; automated constraint checking during animation creation; and designer training in the kinematic vocabulary of nonholonomic systems. Only trajectories that pass feasibility checks should enter user testing, and discrepancies discovered during embodiment should feedback to update the component library and its parameter bounds.

Finally, the evidential base is demographically narrow. Preferences were elicited from a modest cohort of older adults within a single national context, and action-level validation relied on an even smaller convenience sample. To establish generalizability and reveal demographic patterning, future studies should increase sample size, stratify across age groups (including mid-life and younger adults), and diversify cultural contexts. Future studies will include larger, stratified datasets that will enable modeling of individual-difference moderators, technology familiarity, mobility, sensory acuity, and support estimation of continuous preference functions (e.g., speed–comfort curves) rather than binary choices, thereby strengthening both external validity and the precision of the resulting design priors.

References

- Alves-Oliveira, P., Orr, A., Björling, E. A., & Cakmak, M. (2022). Connecting the Dots of Social Robot Design From Interviews With Robot Creators. *Frontiers in Robotics and AI*, 9. <https://doi.org/10.3389/frobt.2022.720799>
- Avijet, P., Aliasghari, P., & Dautenhahn, K. (2024). Greeting Preferences in a Hospitality Context: A Cross-Cultural Study with a Social Robot. *2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)*, 1786–1793. <https://doi.org/10.1109/RO-MAN60168.2024.10731433>
- Bradwell, H. L., Edwards, K. J., Winnington, R., Thill, S., & Jones, R. B. (2019). Companion robots for older people: Importance of user-centred design demonstrated through observations and focus groups comparing preferences of older people and roboticists in South West England. *BMJ Open*, 9(9), e032468. <https://doi.org/10.1136/bmjopen-2019-032468>
- Cass, A. G., Striegnitz, K., & Webb, N. (2018). A farewell to arms: Non-verbal communication for non-humanoid robots. *Proceedings of the Workshop on NLG for Human–Robot Interaction*, 22–26. <https://doi.org/10.18653/v1/W18-6905>
- Castro-González, Á., Admoni, H., & Scassellati, B. (2016). Effects of form and motion on judgments of social robots' animacy, likability, trustworthiness and unpleasantness. *International Journal of Human-Computer Studies*, 90, 27–38. <https://doi.org/10.1016/j.ijhcs.2016.02.004>
- Chiu, C.-J., Hsieh, S., & Li, C.-W. (2021). Needs and Preferences of Middle-Aged and Older Adults in Taiwan for Companion Robots and Pets: Survey Study. *Journal of Medical Internet Research*, 23(6), e23471. <https://doi.org/10.2196/23471>
- Coglan, S., Waycott, J., Lazar, A., & Barbosa Neves, B. (2021). Dignity, Autonomy, and Style of Company: Dimensions Older Adults Consider for Robot Companions. *Proc. ACM Hum.-Comput. Interact.*, 5(CSCW1), 104:1-104:25. <https://doi.org/10.1145/3449178>
- Esterwood, C., Ye, X., Guan, R., & Robert, L. P. (2025). *Virtually The Same or Realistically Different: Do Real-World Robots Differ From Their Non-Physical Representations? :: Data and Code ::* [Dataset]. Zenodo. <https://doi.org/10.5281/ZENODO.14270756>
- Gervasi, R., Aliev, K., Mastrogiacomo, L., & Franceschini, F. (2022). User Experience and Physiological Response in Human-Robot Collaboration: A Preliminary Investigation. *Journal of Intelligent & Robotic Systems*, 106(2), 36. <https://doi.org/10.1007/s10846-022-01744-8>
- Hoffman, G., & Ju, W. (2014). Designing Robots With Movement in Mind. *Journal of Human-Robot Interaction*, 3(1), 89. <https://doi.org/10.5898/JHRI.3.1.Hoffman>
- Hoggenmueller, M., Chen, J., & Hespanhol, L. (2020). Emotional expressions of non-humanoid urban robots: The role of contextual aspects on interpretations. *Proceedings of the 9TH ACM International Symposium on Pervasive Displays*, 87–95. <https://doi.org/10.1145/3393712.3395341>
- Hung, L., Liu, C., Woldum, E., Au-Yeung, A., Berndt, A., Wallsworth, C., Horne, N., Gregorio, M., Mann, J., & Chaudhury, H. (2019). The benefits of and barriers to using a social robot PARO in care settings: A scoping review. *BMC Geriatrics*, 19(1), 232. <https://doi.org/10.1186/s12877-019-1244-6>
- Knight, H., & Simmons, R. (2014). Expressive motion with x, y and theta: Laban Effort Features for mobile robots. *The 23rd IEEE International Symposium on Robot and Human Interactive Communication*, 267–273. <https://doi.org/10.1109/ROMAN.2014.6926264>

- Kruse, T., Pandey, A. K., Alami, R., & Kirsch, A. (2013). Human-aware robot navigation: A survey. *Robotics and Autonomous Systems*, 61(12), 1726–1743. <https://doi.org/10.1016/j.robot.2013.05.007>
- Law, T., de Leeuw, J., & Long, J. H. (2021). How Movements of a Non-Humanoid Robot Affect Emotional Perceptions and Trust. *International Journal of Social Robotics*, 13(8), 1967–1978. <https://doi.org/10.1007/s12369-020-00711-3>
- Macis, D., Perilli, S., & Gena, C. (2022). Employing Socially Assistive Robots in Elderly Care. *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization*, 130–138. <https://doi.org/10.1145/3511047.3537687>
- Olatunji, S. A., Nguyen, V., Cakmak, M., Edsinger, A., Kemp, C. C., Rogers, W. A., & Mahajan, H. P. (2024). Immersive participatory design of assistive robots to support older adults. *Ergonomics*, 67(6), 717–731. <https://doi.org/10.1080/00140139.2024.2312529>
- Rogers, W. A., Kadylak, T., & Bayles, M. A. (2022). Maximizing the Benefits of Participatory Design for Human–Robot Interaction Research With Older Adults. *Human Factors*, 64(3), 441–450. <https://doi.org/10.1177/00187208211037465>
- Saerbeck, M., & Bartneck, C. (2010). Perception of affect elicited by robot motion. *Proceeding of the 5th ACM/IEEE International Conference on Human-Robot Interaction - HRI '10*, 53. <https://doi.org/10.1145/1734454.1734473>
- Sarfraz, D., Ismataellov, U., & Kim, K. (2025). Developing a Hierarchically Structured Action Catalogue for an Elderly Care Robot through Participatory Design. *한국디자인학회 학술발표대회 논문집*, 166–167.
- Sawik, B., Tobis, S., Baum, E., Suwalska, A., Kropińska, S., Stachnik, K., Pérez-Bernabeu, E., Cildoz, M., Agustin, A., & Wieczorowska-Tobis, K. (2023). Robots for Elderly Care: Review, Multi-Criteria Optimization Model and Qualitative Case Study. *Healthcare*, 11(9), 1286. <https://doi.org/10.3390/healthcare11091286>
- Scales, P., Aycard, O., & Aubergé, V. (2024). Planning Socially Expressive Mobile Robot Trajectories. *Sensors*, 24(11), 3533. <https://doi.org/10.3390/s24113533>
- Shibata, T., & Wada, K. (2011). Robot therapy: A new approach for mental healthcare of the elderly - a mini-review. *Gerontology*, 57(4), 378–386. <https://doi.org/10.1159/000319015>
- Sinner, J., & Lim, Y. W. (2021). *Co-designing technology with elders: A systematic review* | *International Journal of Integrated Care*. <https://doi.org/10.5334/ijic.s4042>
- Tsoi, N., Sterneck, R., Zhao, X., & Vázquez, M. (2024). Influence of Simulation and Interactivity on Human Perceptions of a Robot During Navigation Tasks. *ACM Transactions on Human-Robot Interaction*, 13(4), 1–19. <https://doi.org/10.1145/3675784>
- Xie, Q., Chen, Z., & Luh, D. (2025). How Do Humans Recognize the Motion Arousal of Non-Humanoid Robots? *Applied Sciences*, 15(4), 1887. <https://doi.org/10.3390/app15041887>
- Xie, Q., & Luh, D.-B. (2023). The Holistic Expression Factors of Emotional Motion for Non-humanoid Robots. *International Journal of Advanced Computer Science and Applications*, 14(10). <https://doi.org/10.14569/IJACSA.2023.0141068>
- Zhang, M., Xu, T., Hardin, J., Jiaqi Cai, J., Brooks, J., & Green, K. E. (2021). How Many Robots Do You Want? A Cross-Cultural Exploration on User Preference and Perception of an Assistive Multi-Robot System. *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*, 580–585. <https://doi.org/10.1109/RO-MAN50785.2021.9515396>

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